MODULAR CONTRACTS

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FACTS

Current theory: Contract as incentive/sharing rule: outcome $\mapsto$ transfers

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Real contracts

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Real contracts: Written in language and divided into modular clauses

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Evaluated clause by clause

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Evaluated clause by clause $\Rightarrow$ phenomena not in standard model

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<u>Loopholes</u>

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<u>Landmines</u>

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- <u>Conservatism</u>

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<u>Incompleteness</u>: Clauses omit relevant contingencies

<u>Conservatism</u>: High burden of truth to verify clause

QUESTIONS

Why are contracts modular, with clauses evaluated separately?

What explains loopholes, complexity, landmines, incompleteness,...?

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Model of modular contracts

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Premises: Atomic terms

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- Evaluation of clauses: 0 or 1 based on threshold of observed data

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Assumption

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E.g. true data: $\mathbf{a}_j = 110, \mathbf{a}_k = 101$

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Tradeoff:

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Tradeoff: Wide clauses: Susceptible to shuffling; Narrow: Neglect joint info

RESULTS

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Narrow clauses

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Extension: High thresholds

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Extension: High thresholds $\Rightarrow$ conservatism

MODEL

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(Atomic) Premises

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Clause

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Target policy

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Target policy: $\mathcal{T} \subset \mathcal{A}$ evaluated on true data

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MODEL: EXAMPLE

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Premises

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Premises: $\mathbf{a}_{\mathrm{IC}}$ = "interest coverage $\geq \kappa_{\mathrm{IC}}$"; $\mathbf{a}_{\mathrm{LR}}$ = "leverage ratio $\leq \kappa_{\mathrm{LR}}$"

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(Possible) clauses

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Evaluation: $[\![\mathcal{M}_m]\!] = 1$ if holds with sufficient likelihood

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(Possible) clauses: $\mathcal{M}_{\mathrm{IC}} = \{\mathbf{a}_{\mathrm{IC}}\}$, $\mathcal{M}_{\mathrm{LR}} = \{\mathbf{a}_{\mathrm{LR}}\}, \mathcal{M}_{\mathrm{joint}} = \{\mathbf{a}_{\mathrm{IC}}, \mathbf{a}_{\mathrm{LR}}\}$

Contract: $\Phi^{\mathcal{A}} = \{\mathcal{M}_{\mathrm{IC}}, \mathcal{M}_{\mathrm{LR}}\}$ or $\Phi^{\mathcal{T}} = \{\mathcal{M}_{\mathrm{joint}}\}$

Data: $\mathbf{a}_j^i = 1$ iff premise j satisfied in (unobserved) state i

Shuffled data: Know likelihoods premises hold, not state-by-state match

Evaluation: $[\![\mathcal{M}_m]\!] = 1$ if holds with sufficient likelihood

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Cost: Deem contract holds when target violated and vice versa

DEFINITIONS: PREMISE- AND TARGET-BASED

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Target-based contract: $\Phi^{\mathcal{T}} := \left\{\mathcal{T}\right\}$

BENCHMARKS

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In general: Premise- and target-based contracts coincide: $\llbracket \Phi^{\mathcal{A}} \rrbracket = \llbracket \Phi^{\mathcal{T}} \rrbracket$

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Evaluation reduces to logic; grouping of premises irrelevant

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In general: Target-based contract implements target policy: $\llbracket \Phi^{\mathcal{T}} \rrbracket = \llbracket \mathcal{T} \rrbracket$

LOOPHOLES

DEFINITION: LOOPHOLE STATES

Loophole states LH: Data realizations s.t. $[\![\Phi^{\mathcal{A}}]\!] \neq [\![\Phi^{\mathcal{T}}]\!]$ even if $e = 0$

EXAMPLE: LOOPHOLE

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Two premises with three data each and $\tau = 2$:

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Two premises with three data each and $\tau = 2$:

i
1
2
3

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Target-based: $\llbracket \Phi^{\mathcal{T}} \rrbracket = \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket = \llbracket 110 \wedge 101 \rrbracket = \llbracket 100 \rrbracket = 0$

Loophole: Evaluation & conjunction don't commute: $\llbracket \mathbf{a}_j \rrbracket \wedge \llbracket \mathbf{a}_k \rrbracket \neq \llbracket \mathbf{a}_j \wedge \mathbf{a}_k \rrbracket$

LOOPHOLE: CONCRETELY

"Leverage ratio $\leq \kappa_{\text{LR}}$" holds in 2/3 states

"Interest coverage $\geq \kappa_{\text{IC}}$" holds in 2/3 states

Loophole if $\mathbf{a}_{\text{LR}} = 110$ and $\mathbf{a}_{\text{IC}} = 101$:

> High assets (LR holds) & low earnings (IC violated) if spend on R&D
>
> Low assets (LR violated) & high earnings (IC holds) if future earnings low
>
> $\Rightarrow \mathbf{a}_{\text{LR}}$ and $\mathbf{a}_{\text{IC}}$ both hold in 1/3 states

Deem that each term holds but one does not

LOOPHOLE: CONCRETELY

“Interest coverage $\geq \kappa_{\text{IC}}$” $\wedge$ “Leverage ratio $\leq \kappa_{\text{LR}}$” each hold in 2/3 states

Loophole if:

> Earnings low when spending on R&D (IC violated) but long term assets okay (LR holds)
>
> Assets low when future earnings low (LR violated) but current earnings okay (IR holds)

“Interest coverage $\geq \kappa_{\text{IC}} \wedge$ “Leverage ratio $\leq \kappa_{\text{LR}}$” holds in 1/3 states

Deem that each term holds but one does not hold

SHUFFLE RISK

DEFINITION: SHUFFLE RISK STATES

Shuffle risk states SR: Data realizations s.t. $0 < \mathsf{Pr}\left[[\![\Phi^{\mathcal{T}}]\!] = 1 \,\middle|\, \{\mathbf{a}_j\}_j \right] < 1$

EXAMPLE: SHUFFLE RISK

Observe permuted data

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$
1	1	1	1
2	1	0	0
3	0	1	0
# of 1s $\geq \tau$	**1**	**1**	**0**

EXAMPLE: SHUFFLE RISK

Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle)

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1	1	1	1	1
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# of 1s $\geq \tau$	**1**	**1**	**0**	**1**

EXAMPLE: SHUFFLE RISK

Observe permuted data E.g. $\hat{\mathbf{a}}_j = \mathbf{a}_j$ (no shuffle) ; $\hat{\mathbf{a}}_k = 110$ (shuffle of $\mathbf{a}_k$)

i	$\mathbf{a}_j$	$\mathbf{a}_k$	$\mathbf{a}_j \wedge \mathbf{a}_k$	$\hat{\mathbf{a}}_j$	$\hat{\mathbf{a}}_k$
1	1	1	1	1	1
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Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

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1	1	1	1	1	1	1
2	1	0	0	1	1	1
3	0	1	0	0	0	0
# of 1s $\geq \tau$	**1**	**1**	**0**	**1**	**1**	**1**

Marginals preserved $\Rightarrow$ premise-based evaluation unchanged

Joint changed: 101 $\rightarrow$ 110 $\Rightarrow$ target-based evaluation flips

SHUFFLE RISK: CONCRETELY

Back to leverage ratio and interest coverage

True data: $\mathbf{a}_{\mathrm{LR}} = 110$ and $\mathbf{a}_{\mathrm{IC}} = 101$:

High assets go with low earnings

Observations = shuffled data: E.g. $\hat{\mathbf{a}}_{\mathrm{LR}} = 110$ and $\hat{\mathbf{a}}_{\mathrm{IC}} = 110$

Serial correlation: High earnings today $\equiv$ high assets

$\Rightarrow \mathbf{a}_{\mathrm{LR}}$ and $\mathbf{a}_{\mathrm{IC}}$ both hold in 2/3 states

RESULTS

LEMMA

LEMMA: LOOPHOLES ARE FALSE POSITIVES

If $[\![\Phi^{\mathcal{A}}]\!] \neq [\![\mathcal{T}]\!]$, then $[\![\Phi^{\mathcal{A}}]\!] = 1$

LEMMA: LOOPHOLES FALSE POSITIVES: PROOF

Suppose (for contradiction) that $\llbracket \mathcal{T} \rrbracket = 1$ and $\llbracket \Phi^{\mathcal{A}} \rrbracket = 0$

$\llbracket \mathcal{T} \rrbracket = 1 \Rightarrow \bigwedge_{a_j} a_j^i = 1$ for at least τ “rows”

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$\Rightarrow a_j^i = 1$ for at least τ rows for each j

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$$\Rightarrow a_j^i = 1 \text{ for at least } \tau \text{ rows for each } j$$

$$\Rightarrow \llbracket \mathbf{a}_j \rrbracket = 1 \text{ for each } j$$

LEMMA: LOOPHOLES FALSE POSITIVES: PROOF

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$\Rightarrow a_j^i = 1$ for at least τ rows for each j

$\Rightarrow \llbracket \mathbf{a}_j \rrbracket = 1$ for each j

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$\Rightarrow \llbracket \Phi^{\mathcal{A}} \rrbracket = 1$

$\Rightarrow\Leftarrow$

LEMMA

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK

$\mathsf{LH} \subsetneq \mathsf{SR}$

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

LH $\Rightarrow$ SR

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

$\mathsf{LH} \Rightarrow \mathsf{SR}$: Loophole states: Premise-based accepts, target-based rejects

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$\mathsf{LH} \Rightarrow \mathsf{SR}$: Loophole states: Premise-based accepts, target-based rejects

Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

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$\mathsf{LH} \Rightarrow \mathsf{SR}$: Loophole states: Premise-based accepts, target-based rejects

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Target-based accepts too if shuffle "aligns" all 1's

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$\mathsf{SR} \not\Rightarrow \mathsf{LH}$

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$\mathsf{SR} \not\Rightarrow \mathsf{LH}$: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$

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$\mathsf{SR} \not\Rightarrow \mathsf{LH}$: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$: $[\![\mathbf{a}_j \wedge \mathbf{a}_k]\!] = [\![110]\!] = 1$ if $\tau = 2$

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

$\mathsf{LH} \Rightarrow \mathsf{SR}$: Loophole states: Premise-based accepts, target-based rejects

Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

Target-based accepts too if shuffle "aligns" all 1's

$\mathsf{SR} \not\Rightarrow \mathsf{LH}$: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$: $[\![\mathbf{a}_j \wedge \mathbf{a}_k]\!] = [\![110]\!] = 1$ if $\tau = 2$ (not LH)

LEMMA: LOOPHOLES $\subsetneq$ SHUFFLE RISK: PROOF

$\mathsf{LH} \Rightarrow \mathsf{SR}$: Loophole states: Premise-based accepts, target-based rejects

Premise based accepts if $\min_j \sum_i a_j^i \geq \tau$

Target-based accepts too if shuffle "aligns" all 1's

$\mathsf{SR} \not\Rightarrow \mathsf{LH}$: E.g. $\mathbf{a}_j = 110$, $\mathbf{a}_k = 110$: $[\![\mathbf{a}_j \wedge \mathbf{a}_k]\!] = [\![110]\!] = 1$ if $\tau = 2$ (not LH)

But shuffle could be $\hat{\mathbf{a}}_j = 110, \hat{\mathbf{a}}_k = 101$

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But shuffle could be $\hat{\mathbf{a}}_j = 110, \hat{\mathbf{a}}_k = 101$: $[\![\hat{\mathbf{a}}_j \wedge \hat{\mathbf{a}}_k]\!] = [\![100]\!] = 0$ (yes SR)

R1: LOOPHOLES

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With $e > 0$, if $\mathsf{Pr}[\mathsf{SR} \setminus \mathsf{LH}]$ large, then $\Phi^{\mathcal{A}} \succ \Phi^{\mathcal{T}}$

R1: LOOPHOLES: INTUITION

Shuffled data ($e > 0$): Evaluation errors for all but not only loopholes

Accept loopholes to reduce shuffling error

LOOPHOLES IN PRACTICE: SERTA

Serta was distressed and wanted to raise liquidity by priming lenders

Credit agreement ostensibly prohibited non-pro-rata treatment, but…

Clause $\mathbf{a}_j$: "Majority can amend the credit agreement"

Clause $\mathbf{a}_k$: "Amendment can permit open market purchase"

Clause $\mathbf{a}_\ell$: "Selective exchange = open market purchase"

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I.e. loophole: $[\![\mathbf{a}_j]\!] \wedge [\![\mathbf{a}_k]\!] \wedge [\![\mathbf{a}_\ell]\!] \neq [\![\mathbf{a}_j \wedge \mathbf{a}_k \wedge \mathbf{a}_\ell]\!]$

LOOPHOLES IN THE LITERATURE

Katz 10: Many loopholes openly exploited after much litigation

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- Show why loopholes persist $e > 0$

R2: ERRORS DUE TO COMPLEXITY

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If $c_{\mathrm{I}}\mathsf{Pr}[\mathsf{LH}]$ sufficiently high, $\Phi^{\mathcal{T}} \succ \Phi^{\mathcal{A}}$

R2: COMPLEXITY RISK: INTUITION

If loopholes likely or costly use complex target-based contract (long clauses)

Cost: With interacting premises, shuffle risk $\Rightarrow$ likely evaluated with error

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Empirically: Complexity leads to errors in taxes and regulatory compliance
OECD, Benzarti 20, Zwick 21, Behn et al 22...

R3: REDUNDANCY

DEFINITION: REDUNDANCY

$\Phi^{\mathcal{R}}$ called redundant if there's $\mathbf{a}_j$ s.t. $\mathbf{a}_j \in \mathcal{M}_m \cap \mathcal{M}_{m'}$ for clauses $m \neq m'$

R3: REDUNDANCY

If $e > 0$ not too high and $c_{\mathrm{I}}\mathsf{Pr}[\mathsf{LH}]$ high enough, both $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{T}}$ & $\Phi^{\mathcal{R}} \succ \Phi^{\mathcal{A}}$

R3: REDUNDANCY: INTUITION

Benefit w.r.t. target based: Fewer false positives

Repeated clause $\Rightarrow$ require multiple shuffling errors

Benefit w.r.t. premise-based: Fewer loopholes

Keeps joint information

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Keeps joint information

Cost: False negatives: Shuffles in any clause can cause rejection

R4: INCOMPLETENESS

DEFINITION: INCOMPLETENESS

$\Phi^{\mathcal{I}}$ called incomplete w.r.t. $\mathcal{T}$ if $\mathbf{a}_j \notin \bigcup_{\mathcal{M}_m \in \Phi^{\mathcal{I}}} \mathcal{M}_m$ for some $j \in \mathcal{T}$

R4: INCOMPLETENESS

If e and c_{II} high and $\mathsf{Pr}\left[\llbracket\Phi^{\mathcal{I}}\rrbracket > \llbracket\mathcal{T}\rrbracket\right]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

R4: INCOMPLETENESS

If e and c_{II} high and $\mathsf{Pr}\left[[\![\Phi^{\mathcal{I}}]\!] > [\![\mathcal{T}]\!]\right]$ low, then $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{T}}$

"Sometimes" also $\Phi^{\mathcal{I}} \succ \Phi^{A}$

R4: INCOMPLETENESS: INTUITION

Long clauses induce shuffling risk $\Rightarrow$ omit some premises to avoid it

Akin to mismeasurement in Holmström–Milgrom:

Decrease power on distortionary tasks

Could use "proxy task" $\mathbf{a}_\ell \approx \mathbf{a}_j \wedge \mathbf{a}_k$ (yielding $\Phi^{\mathcal{I}} \succ \Phi^{\mathcal{A}}$)

R5: CONSERVATISM

EXTENSION

Clause-dependent thresholds $\tau_{\mathcal{M}}$

DEFINITION: CONSERVATISM

Let $\tau_{\mathcal{T}}$ be threshold of target

$\Phi^{\mathcal{C}}$ is conservative if $\tau_{\mathcal{M}_m} > \tau_{\mathcal{T}}$ for every m

R5: CONSERVATISM

Define $\mathsf{ACC}_\tau(\Phi) := \big\{\{\mathbf{a}_j\}_j \ : [\![\Phi]\!] = 1\big\}$

If $\mathsf{Pr}\left[[\![\mathcal{T}]\!] = 1 \mid \mathsf{ACC}_\tau(\Phi^\mathcal{A}) \setminus \mathsf{ACC}_{\tau+1}(\Phi^\mathcal{A})\right] < \dfrac{c_\mathrm{I}}{c_\mathrm{I} + c_\mathrm{II}}$, then $\Phi^\mathcal{C} \succ \Phi^\mathcal{A}$

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

R5: CONSERVATISM: INTUITION

Premise-based evaluations gives too many false positives

Conservatism decreases them via increased threshold

Feature: Relies on only marginals $\Rightarrow$ not subject to shuffle risk

R5: CONSERVATISM: APPLICATIONS

Law: “beyond reasonable doubt,” “reasonable suspicion,” “probable cause”
In re Winship, 397 U.S. 358 (1970), Terry v. Ohio, 392 U.S. 1 (1968)

Accounting & auditing: High standard for good news Basu 97; Watts 03

E.g. losses recognized faster than gains

CONCLUSION

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Model of modular contracts in formal language s.t. shuffled data

- Loopholes
- Errors of complexity
- Landmines
- Incompleteness
- Conservatism

MODULAR CONTRACTS